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# C Operator Precedence and Associativity

A quick reference for solving expressions correctly

**Rule:** Operators with higher precedence are grouped first. If operators have the same precedence, associativity decides the grouping. Use parentheses whenever they make your intended calculation clearer.

## Complete C precedence sequence - highest to lowest

| Operators | Associativity | What they do |
|---|---|---|
| `() [] -> . postfix ++ postfix --` | Left to right | postfix / access |
| `prefix ++ -- + - ! ~ * & sizeof (type)` | Right to left | unary / cast |
| `* / %` | Left to right | multiply, divide, remainder |
| `+ -` | Left to right | add, subtract |
| `<< >>` | Left to right | bit shifts |
| `< <= > >=` | Left to right | relational comparison |
| `== !=` | Left to right | equality comparison |
| `&  ^  \|` | Left to right | bitwise operations |
| `&&  \|\|` | Left to right | logical AND, logical OR |
| `?:` | Right to left | conditional expression |
| `= += -= *= /= %= &= ^= \|= <<= >>=` | Right to left | assignment |
| `,` | Left to right | comma operator |

## For the present C course, remember this order first

```
() -> unary -> * / % -> + - -> comparisons -> == != -> && -> || -> =
```

## Worked examples

| Expression | Correct grouping and result |
|---|---|
| `result = 2 + 3 * 4;` | `result = 2 + (3 * 4) = 14;`<br>Multiplication has higher precedence than addition. |
| `result = (2 + 3) * 4;` | `result = 5 * 4 = 20;`<br>Parentheses change the grouping. |
| `average = a + b + c / 3.0;` | Only `c` is divided by 3.0. Write `(a + b + c) / 3.0` for the intended average. |

**Best practice:** Do not rely on precedence when a reader may misunderstand the expression. Parentheses are free, clear, and safer.

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